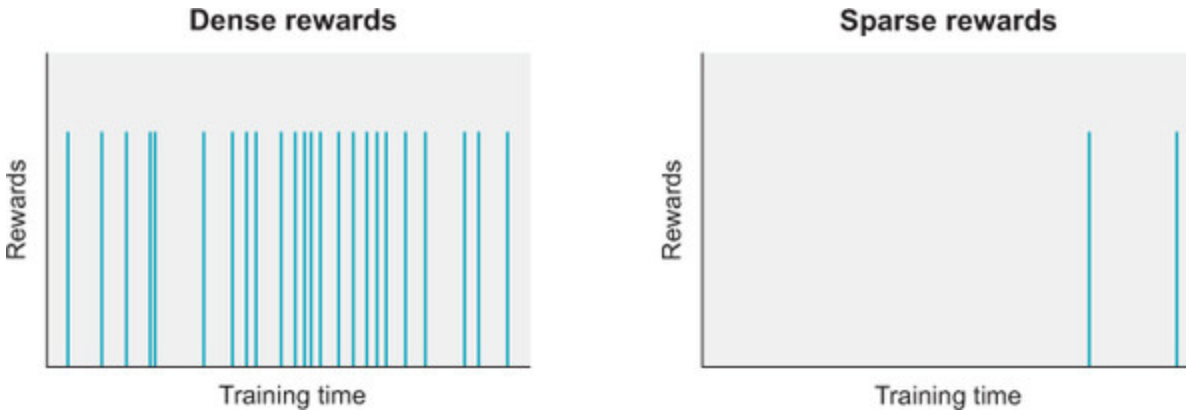
this random exploration policy, so it will never observe a reward and will never learn.

This problem is called the *sparse reward problem*, since the rewards in the environment are sparsely distributed (figure 8.2). If the agent doesn't observe enough reward signals to reinforce its actions, it can't learn.

Figure 8.2. In environments with dense rewards, the rewards are received fairly frequently during the training time, making it easy to reinforce actions. In sparse reward environments, rewards may only be received after many sub-goals are completed, making it difficult or impossible for an agent to learn based on reward signals alone.



Animal and human learning offer us the only natural examples of intelligent systems, and we can turn to them for inspiration. Indeed, researchers trying to tackle this sparse reward problem noticed that humans not only maximize extrinsic rewards (those from the environment), like food and sex, but they also demonstrate an intrinsic curiosity, a motivation to explore just for the sake of understanding how things work and to reduce their uncertainty about their environment.

In this chapter you will we learn about methods for successfully training reinforcement learning agents in sparse reward environments by using principles from human intelligence, specifically our innate curiosity. You will see how curiosity can drive the development of basic skills that the agent can use to accomplish sub-goals and find the sparse rewards. In particular, you will see how a curiosity-powered agent can play the Atari game Super Mario Bros. and learn how to navigate the dynamic terrain just by curiosity alone.